

SMART INDIA HACKATHON 2026

# RAKSHAK K9

Autonomous Multi-Sensor Robotic System for Real-Time Detection of Narcotics and Explosives Across Indian Railways

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**Detailed Technical Project Report — Smart India Hackathon 2026**

Problem Statement ID: SIH26026 | Theme: Robotics and Drones — Category: Hardware | Sponsoring Ministry: Ministry of Railways

*Detailed Technical Project Report*  
Prepared for SIH Technical Jury, Faculty, and Viva Evaluation

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## ABSTRACT

Unattended baggage and person-borne threats involving narcotics and explosives remain a persistent security challenge across Indian Railways' vast, high-footfall, and often unmonitored premises. Existing detection technologies — dual-energy X-ray baggage scanners, Ion Mobility Spectrometry (IMS) swab detectors, Raman spectroscopy scanners, and trained canine units — are effective in principle but are largely fixed-location, operator-dependent, and expensive, making continuous coverage of platforms, yards, and remote sections impractical.

RAKSHAK K9 proposes a wheeled, autonomous, multi-sensor patrol robot that combines low-cost sensing (computer vision, mmWave radar, gas/VOC sensing, and load-cell weight-anomaly detection) with a two-stage, human-verified confirmation process — an on-board UV-fluorescence "black-box" module operated by Railway Protection Force (RPF) personnel. Every stage of the resulting evidence, from first flag to confirmed result, is secured using IPFS content-addressed storage and an Ethereum smart contract, giving Indian Railways a tamper-proof, auditable chain of custody that the project team's market survey found no current commercial system in this space to provide.

This report consolidates the team's research basis, sensor-selection rationale, system architecture, novelty statement, honest limitations, and future scope for RAKSHAK K9, developed in response to Smart India Hackathon 2026 problem statement SIH26026. This document expands the team's original Detailed Project Report into a chapter-structured technical report suitable for jury evaluation, faculty evaluation, and technical demonstration. All technical claims are either drawn directly from the team's original project material, cited to the literature referenced therein, or explicitly flagged where the source material does not provide the required detail. No hardware specification, algorithmic detail, ROS architecture, dataset statistic, or performance figure has been invented for the purpose of this report.

**KEYWORDS:** Autonomous patrol robot, explosives detection, narcotics detection, sensor fusion, YOLOv8, mmWave radar, gas/VOC sensing, UV-fluorescence confirmation, IPFS, blockchain evidence chain, Indian Railways security, Smart India Hackathon.

## ABBREVIATIONS

| Abbreviation | Expansion                                                     |
|--------------|---------------------------------------------------------------|
| SIH          | Smart India Hackathon                                         |
| RPF          | Railway Protection Force                                      |
| CISF         | Central Industrial Security Force                             |
| IMS          | Ion Mobility Spectrometry                                     |
| NQR          | Nuclear Quadrupole Resonance                                  |
| VOC          | Volatile Organic Compound                                     |
| MOX          | Metal-Oxide (gas sensor)                                      |
| UV           | Ultraviolet                                                   |
| NIR          | Near-Infrared                                                 |
| IPFS         | InterPlanetary File System                                    |
| CID          | Content Identifier (IPFS content hash)                        |
| RDX          | Research Department Explosive (cyclotrimethylenetrinitramine) |
| HMX          | High Melting Explosive (cyclotetramethylene-tetranitramine)   |
| PETN         | Pentaerythritol Tetranitrate                                  |
| TNT          | Trinitrotoluene                                               |
| TATP         | Triacetone Triperoxide                                        |
| HMTD         | Hexamethylene Triperoxide Diamine                             |
| GPS          | Global Positioning System                                     |
| YOLO         | You Only Look Once (object-detection model family)            |

| Abbreviation | Expansion                         |
|--------------|-----------------------------------|
| CNN          | Convolutional Neural Network      |
| IoU          | Intersection over Union           |
| mAP          | mean Average Precision            |
| FPS          | Frames Per Second                 |
| API          | Application Programming Interface |
| RF           | Radio Frequency                   |

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## CHAPTER 1 — INTRODUCTION

Indian Railways operates one of the largest rail networks in the world, spanning thousands of stations, platforms, yards, and long stretches of track that are difficult to keep under continuous human observation. Security screening for narcotics and explosives today relies heavily on manual checking by RPF personnel and on fixed-installation equipment concentrated at major stations and checkpoints. This leaves large sections of railway premises — platforms during off-peak hours, yards, sidings, and remote stretches — without active, continuous screening.

RAKSHAK K9 is the team's response to this gap: a wheeled, autonomous, multi-sensor patrol robot intended to provide continuous first-pass screening across railway premises, escalate credible anomalies to RPF personnel through a human-verified on-board confirmation stage, and preserve a cryptographically tamper-evident record of the entire detection-to-confirmation process.

This report is a chapter-structured expansion of the team's original Detailed Project Report. It is written for Smart India Hackathon technical jury evaluation, engineering faculty evaluation, and project viva. The report preserves the original project's technical intent, terminology, and scope in full. Wherever the original material does not specify an implementation detail — for example, exact hardware part numbers beyond those named, communication protocols, or navigation algorithms — this report says so explicitly rather than inventing a plausible-sounding answer. Sections that use general technical/engineering knowledge to explain a referenced technology (for example, how YOLOv8 or Ethereum smart contracts work in general) are clearly marked as general technical background rather than as claims about the team's specific implementation, unless the source material states that detail directly.

## CHAPTER 2 — PROBLEM STATEMENT & REQUIREMENTS

### 2.1 Official Problem Statement

The project responds to SIH26026, issued by the Ministry of Railways under the "Robotics and Drones" theme: "Development of Mobile (Quadruped)/Handheld Device/System for Real-Time Detection of Narcotics and Explosives across Indian Railways." This is the verbatim problem statement as recorded in the source material and has not been altered.

### 2.2 Real-World Operational Challenge

- Indian Railways' scale makes round-the-clock manual coverage of every platform and yard operationally infeasible for RPF personnel alone.
- Fixed-installation detection equipment (X-ray scanners, IMS swab units) is concentrated at major stations and checkpoints, leaving large sections of railway premises without active screening.
- No publicly documented mobile, autonomous system is currently deployed across Indian Railways that combines multi-sensor detection with a legally defensible, tamper-proof evidence trail — this is the specific gap the team identified through its own market survey (Chapter 3).

### 2.3 Why the Problem Requires a Robotic, Multi-Sensor Solution

Manual and fixed-location screening cannot economically scale to continuous coverage. A mobile platform extends screening capability into spaces and time windows that are not otherwise covered. Because, as the team's research found (Chapter 4), no single affordable sensor reliably identifies narcotics or explosives on its own, the robotic platform must carry and fuse multiple independent sensing modalities rather than relying on one.

### 2.4 Human Safety and Inspection Rationale

Continuous automated first-pass screening reduces the exposure of RPF personnel and the travelling public to unscreened threats by widening the coverage envelope, while keeping a human confirmation step (Section 5.2) in the loop for any action with legal or safety consequences, rather than allowing the robot to make an unsupervised final determination.

## 2.5 Design Requirement Derived From the Problem Statement

The problem statement lists "Mobile (Quadruped)" and "Handheld" as example form factors. The team's Detailed Project Report explains its deliberate departure from these examples toward a wheeled platform (Section 2.6 of this chapter, drawn from Section 2.3 of the original report), on engineering grounds specific to the railway environment rather than as a deviation from the problem's intent.

## 2.6 Design Choice: Wheeled Platform

- Railway platforms and yards are predominantly flat, paved surfaces, where wheels perform equally well to legs at a fraction of the mechanical complexity and cost.
- A wheeled chassis is more stable for carrying a multi-sensor payload and offers better power efficiency for extended patrol duty than a legged platform.
- Each wheel strut in the design is servo-actuated, allowing the robot to raise or lower its ground clearance dynamically, providing a degree of terrain adaptability without the cost and complexity of a fully legged system.

[NOT SPECIFIED IN SOURCE MATERIAL: exact wheel count/configuration beyond "4-wheeled", motor type, drivetrain specification, and maximum patrol speed.]

## 2.7 Problem → Technical Challenge → Proposed Engineering Solution

| Problem                                                                                                            | Technical Challenge                                                                   | Proposed Solution                                                                                                          | Expected Benefit                                                                                                              |
|--------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| Continuous screening of vast railway premises is infeasible with manual/fixed checkpoints only                     | Achieving mobile, autonomous, round-the-clock coverage at low per-unit cost           | Wheeled autonomous patrol robot with servo-actuated ground-clearance adjustment                                            | Extends screening coverage to platforms, yards, and remote stretches between fixed checkpoints                                |
| No single affordable sensor reliably identifies narcotics/explosives (non-volatile compounds evade vapour sensing) | Achieving credible detection confidence from low-cost sensors alone                   | Multi-sensor fusion across camera+YOLOv8, mmWave radar, gas/VOC sensor, and load cell                                      | Higher-confidence anomaly flags than any single low-cost sensor, at a fraction of lab-grade equipment cost                    |
| Field sensors cannot safely/definitively confirm hazardous substance identity without contact                      | Need for a safe, low-cost, human-in-the-loop confirmatory test                        | On-board UV-fluorescence "black-box" module operated by RPF after a physical swab                                          | Provides a documented, chemistry-grounded confirmation step before escalation, without requiring lab-grade equipment on-board |
| Conventional databases allow silent alteration or deletion of evidence, weakening legal defensibility              | Need a tamper-evident, auditable chain of custody from first flag to confirmed result | IPFS content-addressed storage (via Pinata with redundant pinning) + append-only Ethereum smart contract (Sepolia testnet) | Cryptographically provable tamper-detection for every stage of the evidence lifecycle                                         |
| Remote yards/tunnels may lack network connectivity                                                                 | Preventing evidence loss during connectivity gaps                                     | Local encrypted buffering with automatic upload on reconnection                                                            | Preserves the evidence trail even when the robot is temporarily offline                                                       |

# CHAPTER 3 — EXISTING SYSTEM (MARKET AND LITERATURE SURVEY)



The team surveyed technologies currently used for explosives and narcotics detection in security, forensic, and humanitarian-demining contexts to understand what already exists, what it costs, and where it falls short for a mobile railway-patrol use case.

### 3.1 Survey of Existing Detection Technologies

| Technology                         | Example Product                                      | Approx. Cost                                          | Key Limitations for This Use Case                                                                                                  |
|------------------------------------|------------------------------------------------------|-------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| Dual-energy X-ray baggage scanning | Smiths Detection HI-SCAN, Rapiscan 620DV             | \$15,000 – \$50,000+                                  | Fixed installation; requires a trained operator to interpret images; cannot patrol or move between locations                       |
| Ion Mobility Spectrometry (IMS)    | Smiths IONSCAN, Rapiscan Itemiser, 908 Devices MX908 | \$30,000 – \$100,000+                                 | Requires a physical swab sample; operator-dependent; not integrated with autonomous mobility or a blockchain-backed evidence trail |
| Raman spectroscopy (handheld)      | Thermo Fisher TruNarc, Rigaku Progeny                | \$20,000 – \$40,000                                   | Needs close/near-contact with a solid sample; no mobility; expensive per unit, limiting fleet-wide deployment                      |
| Nuclear Quadrupole Resonance (NQR) | Quantum Magnetics QScan QR 500                       | Not publicly listed; industrial/institutional pricing | Fixed baggage-screening installation; specialised RF engineering; not available as an off-the-shelf mobile module                  |
| Canine detection units             | Trained K9 units (RPF, CISF)                         | High recurring cost (training, handler, welfare)      | Limited working hours per dog; not scalable to continuous, unmanned coverage of all premises                                       |
| Generic patrol/security robots     | Various commercial surveillance robots               | \$5,000 – \$30,000+                                   | Typically limited to visual surveillance; no integrated explosives/narcotics sensor fusion; no tamper-proof evidence chain         |

Note on cost figures: cost ranges above are approximate figures compiled by the team from publicly available product literature and industry reporting at the time of writing. Exact pricing was not independently verified through a formal manufacturer quote process.

### 3.2 Observed Market Gap

Across every category surveyed, the team found no single commercially available or published system combining all of the following:

- Autonomous, continuous mobile patrol (not fixed-location, not purely handheld);
- Multi-sensor fusion drawing on independent physical detection principles, rather than a single sensing modality;
- A field-deployable, low-cost confirmatory test (rather than only a full laboratory escalation) for suspected explosive residue;
- An immutable, cryptographically verifiable evidence chain (from first detection to human-confirmed result) suitable for legal chain-of-custody use.

This is the specific gap RAKSHAK K9 is designed to fill — not by replacing certified lab-grade equipment (IMS, Raman, NQR), but by providing a continuously patrolling, affordable first-pass screening and evidence-logging layer that flags anomalies for these existing certified systems and RPF personnel to confirm.

## CHAPTER 4 — WHY NO SINGLE AFFORDABLE SENSOR SOLVES THIS PROBLEM

A significant part of the team's research process involved evaluating — and in several cases rejecting — sensor technologies that initially looked promising. This is documented honestly here, as it directly explains the final architecture.

### 4.1 The Core Physical Constraint

Most narcotics and several military-grade explosives (RDX, PETN) are non-volatile: they have extremely low vapour pressure and do not readily evaporate at room temperature [1]. This rules out simple vapour/VOC sensors as a standalone identification method, since what such sensors typically detect is trace solvent or cutting-agent off-gassing rather than the target compound itself.

### 4.2 Sensors Evaluated and Why They Are Insufficient Alone

| Sensor / Technique                                                     | Physical Principle                                           | Reason Insufficient Alone                                                                                                                                                                                                                              |
|------------------------------------------------------------------------|--------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Metal-oxide (MOX) gas sensors (e.g., Bosch BME688)                     | Resistance change on exposure to VOCs                        | Respond to VOCs as a broad class, not a specific compound; cannot reliably detect non-volatile explosives or narcotics                                                                                                                                 |
| Multi-channel visible/NIR spectral sensors (e.g., AS7265x, 410–940 nm) | Reflectance/absorption spectroscopy in visible/NIR band      | Explosives' most distinctive absorption peaks lie in the deep-UV region (RDX ~236 nm, HMX/TNT ~230 nm) [1], outside this sensor's range; NIR patterns alone are not specific enough versus benign organics                                             |
| Deep-UV photodiode + narrow-band UV LED absorption spectroscopy        | Absorption spectroscopy at 230–240 nm                        | Physically sound and correctly targeted, but the required narrow-band deep-UV LEDs are cost-prohibitive (hundreds of dollars per unit) for a student-built prototype                                                                                   |
| Nuclear Quadrupole Resonance (NQR)                                     | RF resonance response of nitrogen-rich crystalline compounds | Most physically robust non-contact technique for non-volatile, nitrogen-rich compounds [5, 6]; even low-cost portable prototypes exist [7], but requires specialised RF pulse-sequencing electronics and RFI shielding beyond a feasible student build |

### 4.3 Conclusion Drawn by the Team

No affordable, single sensing modality reliably identifies these target compounds. The team therefore adopted a sensor-fusion and staged-confirmation architecture instead of searching for one "silver bullet" sensor — an approach the report notes mirrors how real-world security checkpoints operate (metal detector, then X-ray, then trace confirmation), rather than relying on any single machine.

## CHAPTER 5 — PROPOSED SYSTEM

### 5.1 Overview

RAKSHAK K9 is proposed as a two-layer cyber-physical system: an autonomous, continuously operating patrol-and-screening layer, and a human-operated, on-board confirmation layer, with an evidence-integrity layer running underneath both. This structure is a direct design response to the finding in Chapter 4 that no single sensor is sufficient, and to the source material's own framing of the architecture as a deliberate two-layer design rather than a single detection pipeline.

## 5.2 Patrol and Screening Layer

- Camera + YOLOv8 / OpenCV: continuous visual scanning during patrol at low power cost. Only the camera runs during normal patrol; other sensors remain off until an object is flagged. This is a camera-gated sensor-activation design: the camera performs candidate detection, and a flagged object/category triggers activation of the relevant inspection-stage sensors, which then read out and feed the fusion layer. The source material presents reduced power consumption and more targeted screening as the intended benefits of this gating design; this report treats them as design goals rather than measured outcomes, since no power-consumption or false-positive-rate measurements are given (see Chapter 21, Experimental Metrics To Be Measured).
- mmWave radar: used at the inspection stage for close-range, multi-angle scanning to assess shape and density anomalies, complementing the camera's purely visual assessment.
- Gas/VOC sensor (Bosch BME688): must be described precisely as a VOC-sensing / VOC-anomaly-screening input, not as a narcotics or explosives detector. It responds to volatile organic compounds as a broad class; a VOC anomaly does not by itself establish the presence, identity, concentration, or legal classification of a narcotic or explosive substance (Chapter 4.2). It is retained as a broad first-pass screening input despite its lack of compound specificity, because it is inexpensive and still meaningfully raises suspicion confidence for volatile precursor/cutting-agent off-gassing — as one input among several feeding the fusion layer (Chapter 14), never as a standalone identification claim. [CONFIRMATORY DETECTION METHOD: the on-board UV-fluorescence black-box module (Section 5.3) is the specified confirmatory mechanism for suspected explosive residue; no confirmatory chemical mechanism is specified in the source material for narcotics specifically — flagged as a gap in Chapter 26.]
- Load cell + HX711: flags objects that are unusually heavy for their apparent size — a simple, low-cost heuristic also used in manual security screening. This is a weight-anomaly heuristic, not a chemical or material-identification measurement, and should not be described as "detecting" a substance.

## 5.3 Confirmation Layer: On-Board Black-Box Fluorescence Module

Rather than relying on field sensors alone for a final verdict, the design adds a human-operated confirmatory test directly on the robot:

1. RPF personnel collect a physical swab sample of the suspicious residue.
2. A fluorescence-reactive reagent is applied to the swab.
3. The swab is placed inside a sealed on-board chamber containing a UV emitter and a photosensor.
4. UV exposure causes a measurable change in fluorescence intensity if explosive residue is present, which the photosensor records as a confirmatory signal.

This design is grounded in published explosives-detection chemistry rather than being an ad-hoc idea:

- Fluorescence quenching of electron-rich fluorescent materials by electron-poor nitroaromatic explosives (e.g., TNT) is a well-established detection principle, first demonstrated at scale in MIT's FIDO sensor technology [2, 3].
- For non-volatile, nitramine/nitrate-ester explosives such as RDX, HMX, and PETN — exactly the class vapour sensors cannot reliably catch — a UV-triggered fluorescence "fluo-spot" sensor using the DCM fluorophore has been demonstrated in peer-reviewed research, including as a low-cost spray-kit format compatible with naked-eye or photosensor readout [4]. The source material describes this as the closest published analogue to the black-box module's swab-and-spray design.

## 5.4 Why a Two-Layer Design, Not a Single Sensor

- The patrol layer is fully autonomous and continuously operating, maximising coverage.
- The confirmation layer intentionally requires human sample handling, both because current sensor technology cannot safely or reliably identify an unknown hazardous substance without contact, and because a legally defensible chain of custody benefits from a documented human verification step.

## 5.5 Evidence Integrity Layer (Summary)

Every detection event, from the initial autonomous flag through the RPF-confirmed result, is logged as a content-addressed record on IPFS with its hash anchored on an Ethereum smart contract. This layer is detailed in Chapter 22 and is presented here as an architectural pillar of the proposed system rather than an add-on.

## 5.6 System-Level Data Flow (Adapted to the Actual Project)

The generic environment-to-decision pipeline used for robotic threat-detection systems in general has been reconciled below with RAKSHAK K9's actual, documented architecture. Stages not present in the source material (e.g., a dedicated navigation/path-planning subsystem) are marked accordingly rather than assumed.



(Fig. 5.2 — System-Level Data Flow. Adapted from the generic environment-to-decision template to match the two-stage, human-verified architecture actually described in the source material.)

## CHAPTER 6 — SYSTEM ARCHITECTURE

The system is organised into layers below. Each layer's description uses only what the source material specifies; where the source is silent on an interface, protocol, or dependency, this is stated explicitly rather than inferred.

### 6.1 Layer 1 — Physical/Hardware Layer

Purpose: Provide the mobile chassis and payload platform. Inputs: none (ground layer). Processing: mechanical mobility, servo-actuated ground clearance adjustment per wheel strut. Outputs: physical motion and sensor mounting. Technologies: 4-wheeled chassis with servo-actuated wheel struts. Interfaces: [NOT SPECIFIED IN SOURCE MATERIAL — motor driver, motor type, and drivetrain interface not named]. Dependencies: onboard power source [NOT SPECIFIED IN SOURCE MATERIAL — battery type/capacity not given]. Failure conditions: loss of ground clearance actuation on uneven terrain; general mobility failure on non-paved surfaces (ballast, stairs) is explicitly acknowledged in the source material's future-scope discussion (Chapter 26) as outside the current design's target terrain.

### 6.2 Layer 2 — Sensor Acquisition Layer

Purpose: Acquire raw signals from the patrol-layer and inspection-layer sensors. Inputs: physical/environmental stimuli (visual scene, RF reflections, gas concentration, weight). Processing: sensor sampling, per the power-management strategy of running only the camera continuously and activating other sensors on a flag. Outputs: raw camera frames, radar returns, gas-sensor readings, load-cell readings. Technologies: camera, mmWave radar module, Bosch BME688, HX711 load-cell amplifier with load cell. Interfaces: [NOT SPECIFIED IN SOURCE MATERIAL for radar/camera exact interface (e.g., USB, MIPI, SPI); HX711 is a standard digital load-cell amplifier IC, generally interfaced over a simple synchronous serial protocol, stated here as general technical background rather than a confirmed project-specific interface]. Dependencies: Layer 1 for power and mounting. Failure conditions: sensor dropout, occlusion, or false triggering are addressed conceptually in Chapter 20 (Testing) and Chapter 22 (Safety).

### 6.3 Layer 3 — Perception Layer (Computer Vision)

Purpose: Detect candidate objects/anomalies from camera frames using YOLOv8/OpenCV. Inputs: camera frames. Processing: object detection inference (Chapter 10). Outputs: bounding boxes, class labels, confidence scores, triggering the inspection stage. Technologies: YOLOv8 [9], OpenCV. Interfaces: passes detections to the fusion/decision logic. Dependencies: Layer 2 camera feed. Failure conditions: false positives/negatives from the vision model, discussed in Chapter 25 (Limitations).

### 6.4 Layer 4 — Inspection-Stage Sensing Layer

Purpose: Once triggered, gather close-range, multi-modal evidence. Inputs: activation signal from Layer 3. Processing: mmWave radar scanning for shape/density anomaly, gas/VOC concentration sampling, load-cell weight-anomaly check. Outputs: per-sensor readings for fusion. Technologies: as in Section 6.2. Dependencies: Layer 3 trigger. Failure conditions: any individual sensor's known physical limitations (Chapter 4).

### 6.5 Layer 5 — Sensor Fusion Layer

Purpose: Combine independent sensor observations into a single anomaly confidence score. Inputs: outputs of Layer 3 and Layer 4. Processing: weighted sensor-fusion confidence scoring (Chapter 14). Outputs: fused confidence value used to decide whether to escalate to human confirmation. Technologies: [exact fusion software/algorithm implementation: NOT SPECIFIED IN SOURCE MATERIAL beyond "weighted sensor-fusion confidence scoring" as stated in the project's novelty claim]. Dependencies: Layers 3–4. Failure conditions: conflicting or missing sensor readings (Chapter 20).

### 6.6 Layer 6 — Human Confirmation Layer

Purpose: Provide a legally defensible, chemistry-grounded confirmation of suspected explosive residue. Inputs: RPF-collected physical swab. Processing: UV-fluorescence reaction inside the sealed black-box chamber; photosensor readout. Outputs: confirmed/not-confirmed result. Technologies: UV emitter, photosensor, fluorescence-reactive reagent, sealed chamber. Dependencies: Layer 5 escalation and RPF personnel action. Failure conditions: dependent on

correct reagent application and human procedure; the source material does not provide a false-positive/false-negative rate for this stage.

## 6.7 Layer 7 — Evidence Integrity Layer

Purpose: Create a tamper-evident record of every detection and confirmation event. Inputs: sensor readings, images, GPS, timestamps, confirmation results. Processing: IPFS content-addressed storage (via Pinata plus a secondary independent pinning provider) and an append-only Ethereum smart contract (deployed on the Sepolia testnet for the prototype) storing the CID and metadata. Outputs: an "unconfirmed" evidence package at first flag and a "confirmed" evidence package after RPF verification, cryptographically linked. Dependencies: network connectivity, with local encrypted buffering and automatic upload on reconnection when connectivity is unavailable. Failure conditions: the source material explicitly acknowledges that the design guarantees tamper detection but not tamper recovery — if all redundant copies of a file were destroyed simultaneously, the content cannot be reconstructed from the hash alone.

## 6.8 Layer 8 — Communication/Monitoring Layer

Purpose: Move data between sensors, processing, the evidence layer, and RPF/monitoring personnel. [Exact communication technology (Wi-Fi/cellular/other), dashboard software, and monitoring interface: NOT SPECIFIED IN SOURCE MATERIAL beyond the described offline-buffering behaviour.] The team's future-scope section (Chapter 26) references a "shared control dashboard" for future multi-robot fleet coordination, which the source material presents as future work rather than a current capability.

## 6.9 Diagrams Referenced in This Chapter

Fig. 6.1 depicts the eight layers above stacked from physical hardware (bottom) to communication/monitoring (top), matching Section 6.1–6.8, with the Evidence Integrity Layer (6.7) drawn alongside the Sensor Fusion and Human Confirmation layers to reflect that it consumes outputs from both rather than sitting strictly above them.

# CHAPTER 7 — HARDWARE ARCHITECTURE

## 7.1 Component Table

| Hardware                      | Function                                                                        | Data Generated                          | Interface                                                                                                                                                                        | Role                                                    |
|-------------------------------|---------------------------------------------------------------------------------|-----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|
| Camera                        | Continuous visual patrol monitoring; input to YOLOv8/OpenCV detection           | Image frames                            | Specification not provided in source material                                                                                                                                    | Patrol-layer perception (primary, always-on sensor)     |
| mmWave radar module           | Close-range, multi-angle scanning for shape/density anomaly at inspection stage | Radar returns / range-reflectivity data | Specification not provided in source material                                                                                                                                    | Inspection-layer sensing, complements camera            |
| Bosch BME688 (gas/VOC sensor) | Broad-class VOC/gas sensing for volatile precursor/cutting-agent off-gassing    | Gas concentration / VOC index readings  | Specification not provided in source material (commonly interfaced via I2C/SPI per manufacturer datasheet — general technical background, not confirmed project-specific detail) | Inspection-layer sensing (non-specific screening input) |
| Load cell + HX711 amplifier   | Detects objects unusually heavy for their apparent size                         | Weight reading                          | HX711 is a dedicated ADC/amplifier IC for load cells (general technical background)                                                                                              | Inspection-layer sensing (weight-anomaly heuristic)     |

| Hardware                                    | Function                                                          | Data Generated                | Interface                                     | Role                             |
|---------------------------------------------|-------------------------------------------------------------------|-------------------------------|-----------------------------------------------|----------------------------------|
| UV emitter + photosensor (black-box module) | Confirmatory UV-fluorescence readout of RPF-collected swab sample | Fluorescence intensity signal | Specification not provided in source material | Human confirmation layer         |
| Servo-actuated wheel struts                 | Dynamic ground-clearance adjustment for terrain adaptability      | Actuator position/feedback    | Specification not provided in source material | Physical/hardware layer mobility |
| 4-wheeled chassis                           | Structural platform and mobility base                             | N/A                           | N/A                                           | Physical/hardware layer          |

[NOT SPECIFIED IN SOURCE MATERIAL: compute unit / single-board computer or microcontroller model, motor driver, battery type and capacity, GPS module make/model (a GPS-tagged timestamp is referenced as part of the evidence package in Chapter 22, implying a GPS source exists, but no module is named), and communication module make/model.]

## 7.2 Hardware Selection Rationale

The wheeled configuration and servo-actuated struts were chosen over a legged/quadruped configuration for reasons of mechanical simplicity, payload stability, and power efficiency appropriate to predominantly flat, paved railway premises (Section 2.6). Each sensor in Section 7.1 was selected specifically because it is inexpensive relative to lab-grade detection equipment (Chapter 3), and its individual insufficiency (Chapter 4) is compensated by fusing it with the others rather than by upgrading any single sensor.

# CHAPTER 8 — SOFTWARE ARCHITECTURE

## 8.1 Software / Technology Stack

| Layer                        | Technology                                                                                                                         | Purpose                                                                | Status per Source Material                                                                     |
|------------------------------|------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| Computer vision              | YOLOv8 (Ultralytics) [9]                                                                                                           | Real-time object detection during patrol                               | Named directly in source as the patrol-layer detection framework                               |
| Computer vision (supporting) | OpenCV                                                                                                                             | Image processing supporting the vision pipeline                        | Named directly in source                                                                       |
| Evidence storage             | IPFS, accessed via Pinata (primary) plus a secondary independent pinning provider                                                  | Content-addressed, tamper-evident storage of evidence files            | Named directly in source; described as the prototype's storage design                          |
| Evidence integrity ledger    | Ethereum smart contract, deployed on the Sepolia testnet                                                                           | Append-only on-chain storage of evidence CIDs and metadata             | Named directly in source as the current prototype's deployment target                          |
| Fusion logic                 | Weighted sensor-fusion confidence scoring                                                                                          | Combine independent sensor observations into a single confidence value | Named directly in source as a core design element; exact software implementation not specified |
| Local resilience             | Encrypted local buffering with automatic upload on reconnection                                                                    | Preserve evidence during network outages                               | Named directly in source                                                                       |
| [Not specified]              | Operating system, programming languages, robot middleware, dashboard/monitoring software, database (if any) beyond IPFS/blockchain | —                                                                      | Not specified in source material                                                               |

## 8.2 Discussion

YOLOv8 is described in the source material as "an actively maintained, well-benchmarked open object-detection framework" [9], used for the patrol layer's continuous visual scanning. The evidence-integrity software stack (IPFS + Ethereum smart contract) is the most concretely specified part of the software architecture in the source material, including the specific testnet (Sepolia) used for the prototype deployment — this is treated in this report as the closest thing to a confirmed "implemented" software component, though the source material does not state whether this deployment has been exercised end-to-end with the physical sensor hardware or only as a standalone blockchain/IPFS integration. This distinction is called out explicitly in Chapter 20 (Implementation Status).

## CHAPTER 9 — ROS/ROS 2 ARCHITECTURE

The source material does not mention the Robot Operating System (ROS or ROS 2), any ROS nodes, topics, services, actions, TF frames, or a navigation stack, at any point. Per this report's instructions, no ROS architecture, computational graph, node/topic table, or ROS-based navigation stack has been fabricated for RAKSHAK K9.

| Item                          | Status                                                                    |
|-------------------------------|---------------------------------------------------------------------------|
| ROS/ROS 2 use                 | Not specified in source material — no evidence the project uses ROS/ROS 2 |
| Node/topic architecture       | Not applicable — no nodes or topics are named in the source material      |
| Navigation stack (e.g., Nav2) | Not specified in source material                                          |
| TF/TF2 coordinate frames      | Not specified in source material                                          |

If the team is in fact using ROS/ROS 2 internally, this chapter should be revised with the actual node graph, topic names, and message types once that information is provided; this report has intentionally left it as a gap rather than guessing at plausible node names.

## CHAPTER 10 — COMPUTER VISION & YOLOv8

### 10.1 Role in the Project

YOLOv8 and OpenCV form the patrol layer's continuous visual-scanning pipeline (Section 5.2), running as the only always-on sensor during normal patrol to keep power consumption low, with detections triggering the inspection-stage sensors.

### 10.2 General Technical Background: How YOLOv8-Family Detectors Work

The following is general technical explanation of the YOLO object-detection approach, provided as background rather than as a claim about RAKSHAK K9's specific model configuration, training data, or measured accuracy — none of which are specified in the source material.

A YOLO-family detector is a single-stage convolutional neural network (CNN) that predicts object locations and classes in one forward pass, rather than first proposing regions and then classifying them separately. Conceptually:

```
Camera --> Image acquisition --> Preprocessing (resize/normalise) --> CNN backbone
--> Neck (multi-scale feature aggregation) --> Detection head
--> Bounding-box regression + class prediction + confidence score
--> Non-Maximum Suppression (NMS) --> Final detections --> Downstream fusion/decision logic
```

- **Backbone:** a CNN stack that extracts hierarchical visual features from the input image.
- **Neck:** aggregates features across multiple scales so both small and large objects can be detected.
- **Detection head:** predicts, for each candidate location, a bounding box, an objectness/confidence score, and a class probability distribution.
- **Bounding-box regression:** the network predicts offsets that refine a candidate box to tightly enclose the object.



- Non-Maximum Suppression (NMS): removes duplicate overlapping boxes for the same object, keeping the highest-confidence box and suppressing others above an IoU overlap threshold.
- Intersection over Union (IoU): a standard metric for how well a predicted box overlaps a ground-truth box, defined in Chapter 18.

### 10.3 Training, Validation, Testing, and Deployment (General Background)

A general supervised object-detection workflow — labelling images with bounding boxes and class names, splitting data into training/validation/test sets, training the network, tuning on the validation set, evaluating on a held-out test set, and exporting the trained model for on-device inference — is standard practice for deploying a YOLO-family model. [DETAIL REQUIRED FROM TEAM: dataset source, number of images, class list, annotation method, augmentation strategy, train/validation/test split, number of training epochs, and any measured precision/recall/F1/IoU/mAP values are NOT SPECIFIED IN SOURCE MATERIAL and have not been fabricated for this report.] The source material's future-scope section (Chapter 26) states an intention to expand training data by partnering with authorised agencies, using legal simulants during prototype development — implying that training has used simulant-based imagery, though no dataset details are given.

#### 10.3a Dataset Category Breakdown

Datasets serving different purposes must not be conflated. A railway-platform image dataset, for instance, trains a model to recognise the railway environment and general objects — it does not by itself train a narcotics/explosives detector unless it specifically contains labelled examples of the target classes. The table below records what is and is not established by the source material.

| Dataset Category                                                                        | Purpose                                                                         | Status per Source Material                                                                                                                                                                                                                                                        |
|-----------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A. Railway/environment perception data                                                  | Recognising the platform/yard environment and general objects for patrol vision | [NOT SPECIFIED IN SOURCE MATERIAL]                                                                                                                                                                                                                                                |
| B. Object/threat detection data (labelled narcotics/explosives-like objects or proxies) | Training YOLOv8 to flag candidate threat objects                                | Source material states future training-data expansion will use data from "authorised agencies," implying current training (if any) uses legal simulants rather than real controlled substances (Chapter 27); exact current dataset composition [NOT SPECIFIED IN SOURCE MATERIAL] |
| C. Sensor datasets (radar/gas/load-cell reference data)                                 | Calibrating fusion-layer confidence per sensor                                  | [NOT SPECIFIED IN SOURCE MATERIAL]                                                                                                                                                                                                                                                |
| D. Synthetic/generated data                                                             | Augmenting limited real-world samples                                           | [NOT SPECIFIED IN SOURCE MATERIAL]                                                                                                                                                                                                                                                |
| E. Experimental field data (from actual test runs)                                      | Measuring real-world performance (Chapter 21)                                   | [NOT SPECIFIED IN SOURCE MATERIAL — no field-test data reported]                                                                                                                                                                                                                  |

No claim is made in this report that any dataset used contains validated narcotics/explosives imagery with proper annotation, since the source material does not state this.

### 10.4 Metrics Defined for Future Reporting

The following standard computer-vision metrics are defined here so the team can populate them once measured; no values are asserted:

- Precision = True Positives / (True Positives + False Positives)
- Recall = True Positives / (True Positives + False Negatives)
- F1 Score =  $2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$
- IoU = Area of Intersection / Area of Union (Chapter 18)

- mAP@0.5 = mean Average Precision at an IoU threshold of 0.5
- mAP@0.5:0.95 = mean Average Precision averaged over IoU thresholds from 0.5 to 0.95

## CHAPTER 11 — mmWAVE RADAR & SPATIAL PERCEPTION

No LiDAR is referenced anywhere in the source material; the project's spatial/ranging sensor at the inspection stage is a mmWave radar module. This chapter has been adapted accordingly rather than describing a sensor the project does not use.

### 11.1 Role in the Project

mmWave radar is used at the inspection stage, once the camera has flagged a candidate object, for close-range, multi-angle scanning to assess shape and density anomalies — complementing the camera's purely visual assessment (Section 5.2).

### 11.2 General Technical Background: mmWave Radar Principle

This subsection explains the general operating principle of mmWave radar as a technology class, not a confirmed specification of the module used by the project. A mmWave radar transmits frequency-modulated continuous-wave (FMCW) millimetre-wave signals and analyses the reflected returns' time delay, frequency shift (Doppler), and phase across multiple receive antennas to estimate the range, velocity, and angle of reflecting objects. Because radar reflectivity depends on material dielectric properties and geometry, it can indicate the presence of a dense or unusually shaped object inside a bag or package without requiring the object to be visually exposed, and can operate independently of ambient lighting conditions — a general property of mmWave radar noted here as background, not as a measured project result.

### 11.3 Conceptual Pipeline

```
mmWave radar --> RF return signals --> Range/Doppler/angle processing
--> Shape/density anomaly indicators --> Fusion layer (Chapter 14)
```

### 11.4 Limitations Relevant to This Sensor Class

General limitations of mmWave radar as a technology (background, not confirmed field-tested results for this project) include reduced angular resolution compared to optical sensors, potential ambiguity between multiple closely spaced objects, and sensitivity to strong RF-reflective or RF-absorptive materials. [Exact radar module specifications, range, and field-of-view: NOT SPECIFIED IN SOURCE MATERIAL.]

## CHAPTER 12 — LOCALIZATION, MAPPING & NAVIGATION

The source material does not describe a localization, mapping, or path-planning subsystem, SLAM implementation, or navigation stack for RAKSHAK K9. It states only that the robot is a wheeled, autonomous patrol platform with servo-actuated wheel struts for dynamic ground-clearance adjustment (Section 2.6). Per this report's instructions, no odometry, IMU, SLAM, global/local planner, or costmap architecture has been invented.

| Item                                         | Status                                                                                                                                                                          |
|----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Odometry                                     | Not specified in source material                                                                                                                                                |
| IMU                                          | Not specified in source material                                                                                                                                                |
| SLAM / mapping                               | Not specified in source material                                                                                                                                                |
| Path planning / obstacle avoidance algorithm | Not specified in source material                                                                                                                                                |
| GPS                                          | A GPS-tagged timestamp is referenced as part of the logged evidence package (Chapter 22), implying some GPS capability exists, but no module or accuracy specification is given |

| Item                       | Status                                                                                                                                    |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Ground-clearance actuation | Specified: servo-actuated wheel struts allow the robot to raise/lower ground clearance dynamically for terrain adaptability (Section 2.6) |

This is flagged as a priority area for the team to document further, since autonomous "patrol" inherently implies some navigation approach (pre-programmed routes, teleoperation, or autonomous path planning), and the jury may reasonably ask which of these RAKSHAK K9 currently uses.

## 12.1 GPS Availability and Limitations

GPS is referenced only implicitly, via a GPS-tagged timestamp included in the logged evidence package (Chapter 22). The source material does not describe GPS's role beyond this, its expected accuracy, or its behaviour when unavailable. This report adds the following as general engineering caution rather than as a project-specific claim: GPS reception is typically degraded or unavailable inside stations, tunnels, underpasses, and near large steel structures such as railway yards and rolling stock, which are precisely the environments this project targets. [NOT SPECIFIED IN SOURCE MATERIAL: fallback location method when GPS is unavailable — e.g., last-known-position, station/zone ID, or odometry-based estimate.] Continuous GPS availability should not be assumed for any evidence record generated indoors or in an obstructed railway environment, and the team should specify a fallback location-tagging method before field deployment.

# CHAPTER 13 — SENSOR DATA ACQUISITION & PROCESSING

## 13.1 Conceptual Pipeline

```
Sensor (camera / radar / gas / load cell) --> Raw signal --> Sensor-specific acquisition
--> Timestamping (and GPS tagging, per Chapter 22) --> [Calibration/filtering: not specified]
--> Per-sensor confidence/anomaly indicator --> Fusion layer (Chapter 14)
```

## 13.2 What Is and Is Not Specified

The source material establishes that: (a) the camera runs continuously while other sensors are activated only after a visual flag, as a deliberate power-management design; (b) evidence records include images, sensor readings, GPS, and timestamps (Chapter 22). It does not specify calibration procedures, filtering/denoising algorithms, sampling rates, or a synchronisation mechanism across sensors. These are marked [NOT SPECIFIED IN SOURCE MATERIAL] rather than assumed, since inventing a specific filtering algorithm (e.g., a particular Kalman filter variant) would misrepresent the project's documented design.

# CHAPTER 14 — MULTI-SENSOR FUSION

This is one of the project's core technical contributions as documented in the source material, and is treated here as the strongest technical chapter, consistent with the emphasis the team itself places on it in its novelty claim (Chapter 24).

## 14.1 Why a Single Sensor Is Insufficient

As established in Chapter 4, no affordable single sensor reliably identifies narcotics or explosives. Each sensor answers a different, partial question:

- Camera (YOLOv8): "What does the object look like, and is it visually anomalous?"
- mmWave radar: "What is the object's shape and density, independent of visual appearance?"
- Gas/VOC sensor: "Is there an abnormal volatile/off-gassing signature nearby?"
- Load cell: "Is this object unusually heavy for its apparent size?"

### 14.1a Subsystem Evidence Taxonomy (Do Not Merge Into a Single "Detects Explosives" Claim)

Each subsystem produces a distinct category of evidence. These categories must not be collapsed into a single claim such as "the robot detects explosives"; the table below states, for each category, exactly what is and is not established.

| Category                           | Subsystem                                               | What It Establishes                                                                                                                         | What It Does NOT Establish                                                                                             |
|------------------------------------|---------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| A. Visual/object detection         | Camera + YOLOv8                                         | A candidate object/region of visual interest and a class/confidence score                                                                   | Chemical identity, composition, or legal classification of any substance                                               |
| B. Spatial detection/localization  | mmWave radar                                            | Shape and density anomaly of the candidate object at close range                                                                            | Chemical identity of the object's contents                                                                             |
| C. Metal detection                 | Not present in source material                          | N/A                                                                                                                                         | [NOT SPECIFIED IN SOURCE MATERIAL — no metal detector is named]                                                        |
| D. VOC anomaly detection           | Bosch BME688                                            | Presence of an elevated/anomalous VOC signature in the vicinity                                                                             | Which compound is present, its concentration, or whether it is a controlled/hazardous substance                        |
| E. Confirmatory chemical detection | On-board UV-fluorescence black-box module (Section 5.3) | A fluorescence-based, chemistry-grounded indication consistent with certain nitro-compound classes [2,3,4], following human swab collection | A full forensic/legal-grade identification equivalent to a laboratory instrument (IMS/Raman/NQR, Chapter 3)            |
| F. Final threat classification     | RPF personnel (human-in-the-loop), informed by A–E      | The operational decision to treat an item as confirmed/not confirmed and to escalate                                                        | An automated, sensor-only "guilty" determination — the design keeps this decision with a human operator (Section 15.3) |

### 14.2 Fusion Strategy as Documented

The source material describes the fusion approach as "weighted sensor-fusion confidence scoring across independent physical detection principles," presented as a proposed/core design element of the system rather than as a fully specified, tested algorithm with fixed weights. This report therefore labels the mathematical treatment below as the Proposed Sensor Fusion Strategy, not as an already-validated implementation.

### 14.3 Proposed Sensor Fusion Strategy (Conceptual Weighted Model)

A generic weighted-confidence fusion model, consistent with the "weighted sensor-fusion confidence scoring" description in the source material, can be expressed as:

$$C_{fused} = \sum (w_i \times C_i)$$

where  $C_i$  is the confidence/anomaly indicator contributed by sensor  $i$ , and  $w_i$  is the reliability weight assigned to sensor  $i$ , subject to  $\sum w_i = 1$ . This is decision-level fusion: each sensor first produces its own partial confidence/anomaly indicator, and these are then combined, rather than raw signals being merged before independent interpretation (feature-level or data-level fusion). [Exact weight values, calibration method, and whether fusion is implemented as decision-level, feature-level, or data-level in the team's actual code: NOT SPECIFIED IN SOURCE MATERIAL — the classification above as decision-level is the report's engineering interpretation of the "confidence scoring" description, not a verbatim project claim.]

### 14.4 Handling Contradictory or Missing Sensor Data

The source material does not specify a rule for resolving contradictory sensor readings (e.g., strong visual anomaly but no gas or radar corroboration) or for handling a failed/offline sensor within the fusion calculation. This is identified as a gap for the team to formalise, and is listed again in Chapter 25 (Limitations) and Chapter 27 (Future Scope candidate).

## 14.5 Sensor-Level Contribution Summary

| Sensor                                         | Question Answered                                           | Specificity to Target Compounds                                                                                                | Role in Fusion                                                            |
|------------------------------------------------|-------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|
| Camera + YOLOv8                                | Visual anomaly / object identity                            | Low-to-moderate (visual proxies only, cannot chemically confirm)                                                               | Primary trigger for escalation                                            |
| mmWave radar                                   | Shape / density anomaly                                     | Moderate (geometric/dielectric proxy, not chemical)                                                                            | Corroborating evidence at inspection stage                                |
| Gas/VOC sensor (BME688)                        | Off-gassing / volatile signature                            | Low (broad-class VOC response, not compound-specific — Chapter 4)                                                              | Supplementary suspicion-raising input                                     |
| Load cell + HX711                              | Weight-to-size anomaly                                      | Low (heuristic, not detection-specific)                                                                                        | Supplementary suspicion-raising input                                     |
| UV-fluorescence black-box (confirmation layer) | Chemistry-specific fluorescence response to nitro-compounds | High for the documented chemistry (nitroaromatics via quenching [2,3]; nitramine/nitrate esters via fluoro-spot chemistry [4]) | Final human-verified confirmation, not part of the automated fusion score |

## CHAPTER 15 — THREAT CLASSIFICATION & DECISION-MAKING

### 15.1 Decision Logic as Documented

```

Camera flags candidate object
  |
  v
Inspection-stage sensors activated (radar, gas, load cell)
  |
  v
Sensor readings fused into an anomaly confidence value (Chapter 14)
  |
  v
Confidence sufficient to warrant human attention?
  [exact threshold: NOT SPECIFIED IN SOURCE MATERIAL]
  |
  YES                                NO --> continue autonomous patrol
  |
  v
RPF alerted; physical swab collected
  |
  v
On-board UV-fluorescence confirmation (Section 5.3)
  |
  v
Confirmed / Not Confirmed (human-verified) --> Evidence logged (Chapter 22) --> RPF response

```

### 15.2 Conceptual Threat Decision Matrix

The following matrix is a conceptual illustration of how independent sensor evidence might combine to shape the escalation decision. It is explicitly labelled conceptual because the source material does not provide the specific decision rules or thresholds implemented in code.

| Visual (YOLOv8) | Radar Evidence | Gas Evidence | Weight Evidence | Illustrative Decision (conceptual)                                                       |
|-----------------|----------------|--------------|-----------------|------------------------------------------------------------------------------------------|
| High            | High           | High         | High            | Escalate immediately to RPF for confirmation                                             |
| High            | High           | Low          | Low/Neutral     | Escalate — shape/visual anomaly corroborated even without gas/weight signal              |
| Low             | Low            | High         | Low             | Treat as environmental/background anomaly; log but do not necessarily escalate           |
| Low             | Low            | Low          | High            | Weight anomaly alone; flag for lower-priority manual check                               |
| High            | Low            | Low          | Low             | Visual anomaly only; flag for closer camera-only re-inspection before radar/gas re-check |

This matrix is the report's own illustrative construction, built strictly from the four sensor types the source material names, to make the fusion concept concrete for jury evaluation; it is not a reproduction of an internal decision table from the project, since none was provided.

### 15.3 Human-in-the-Loop Principle

At every point where an action would have legal or safety consequences (alerting RPF, treating a sample as confirmed explosive residue, logging a "confirmed" evidence record), the documented design keeps a human decision-maker (RPF personnel) in the loop rather than allowing the robot to act autonomously on a confirmed-threat basis. This is presented in the source material as a deliberate choice tied to legal defensibility, not merely a technical limitation.

### 15.4 Autonomous Functions vs. Operator-Assisted Functions

To avoid implying full autonomy where the design is intentionally human-in-the-loop, functions are separated explicitly below.

| Function                                            | Classification                      | Basis                                                                                                                      |
|-----------------------------------------------------|-------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| Continuous patrol visual scanning (camera + YOLOv8) | Autonomous                          | Described as continuously running without operator input during normal patrol                                              |
| Inspection-stage sensor activation on a visual flag | Autonomous (gated by camera output) | Described as triggered automatically once the camera flags a candidate object                                              |
| Multi-sensor confidence fusion                      | Autonomous                          | Described as an automated scoring step prior to any human action                                                           |
| Ground-clearance adjustment (servo-actuated struts) | Autonomous (per Section 2.6)        | Described as a dynamic, presumably automatic mechanical adjustment; exact trigger logic [NOT SPECIFIED IN SOURCE MATERIAL] |
| Patrol route / path selection                       | [NOT SPECIFIED IN SOURCE MATERIAL]  | Not described — could be pre-programmed, teleoperated, or autonomous; team confirmation required                           |
| Physical swab collection                            | Operator-assisted (RPF personnel)   | Explicitly described as an RPF action                                                                                      |

| Function                                      | Classification                                           | Basis                                                                           |
|-----------------------------------------------|----------------------------------------------------------|---------------------------------------------------------------------------------|
| Confirmatory fluorescence test execution      | Operator-assisted (RPF loads swab into black-box module) | Explicitly described as requiring human sample handling by design (Section 5.4) |
| Final "confirmed/not confirmed" determination | Operator-assisted (human-verified)                       | Explicitly described as RPF-verified                                            |
| Escalation / incident response decision       | Operator-assisted (RPF)                                  | Outside the robot's automated scope, per the source material                    |

## CHAPTER 16 — ROBOT CONTROL

The source material specifies servo-actuated wheel struts for dynamic ground-clearance adjustment (Section 2.6) but does not specify the drive kinematics (e.g., differential drive, Ackermann steering, or independent four-wheel drive), motor type, or closed-loop control scheme.

| Item                                                                       | Status                                                                                           |
|----------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| Drive configuration                                                        | 4-wheeled; exact steering/drive kinematics not specified in source material                      |
| Ground-clearance control                                                   | Servo-actuated per wheel strut, allowing dynamic raising/lowering — specified in source material |
| Closed-loop motor control / encoder feedback                               | Not specified in source material                                                                 |
| Patrol path control (pre-programmed route vs. teleoperated vs. autonomous) | Not specified in source material                                                                 |

No kinematic equations (e.g., differential-drive  $v/\omega$  relationships) have been included here, since applying them would assume a drivetrain configuration the source material does not confirm.

## CHAPTER 17 — COMMUNICATION ARCHITECTURE

### 17.1 Documented Communication Behaviour

- If network connectivity is unavailable (e.g., remote yards, tunnels), evidence is encrypted and buffered locally, then automatically uploaded once connectivity resumes. This is the most concrete communication-reliability behaviour specified in the source material.
- Evidence storage communication runs through Pinata (an IPFS pinning service) as the primary path, with a secondary independent pinning provider for redundancy.
- The blockchain layer communicates with the Ethereum Sepolia testnet for the prototype.

### 17.2 What Is Not Specified

The physical communication medium between the robot and any base station or dashboard (Wi-Fi, cellular/4G-5G, LoRa, or other), telemetry format, command-transmission protocol for any remote/teleoperation link, and latency figures are not given in the source material. A "shared control dashboard" is mentioned only as future-scope for multi-robot fleet coordination (Chapter 26), not as an existing capability.

## CHAPTER 18 — END-TO-END SYSTEM WORKFLOW

The following operational scenario is constructed directly from the architecture described across Chapters 5–17, sequenced as an illustrative patrol-to-evidence cycle. Steps not evidenced in the source material are marked.

1. Robot begins autonomous patrol of an assigned railway platform/yard section. [Route determination method: not specified in source material.]

2. Camera runs continuously; YOLOv8/OpenCV performs real-time visual scanning. All other sensors remain powered down for energy efficiency.
3. YOLOv8 flags a candidate object/anomaly in the visual feed.
4. Inspection-stage sensors (mmWave radar, BME688 gas sensor, HX711 load cell) power on and take readings of the flagged object.
5. Sensor readings are timestamped and GPS-tagged.
6. Readings are fused into a weighted anomaly confidence score (Chapter 14).
7. If the fused confidence is sufficient [threshold not specified], the robot alerts RPF personnel and generates an "unconfirmed" evidence package (images, sensor readings, GPS, timestamp), stored on IPFS with its CID anchored on the Ethereum smart contract.
8. RPF personnel respond, collect a physical swab of the suspicious residue, and apply the fluorescence-reactive reagent.
9. The swab is inserted into the on-board black-box module; the UV emitter and photosensor record the fluorescence response.
10. A confirmed / not-confirmed determination is made based on the fluorescence readout, verified by RPF personnel.
11. A "confirmed" evidence package is generated, referencing the original unconfirmed record's CID/transaction ID, preserving the full investigation timeline immutably.
12. If network connectivity was unavailable at any stage, evidence was encrypted and buffered locally, then uploaded automatically once connectivity resumed.
13. RPF proceeds with incident response per standard operating procedure. [Downstream incident-response procedure itself is outside the scope of the source material.]
14. The robot resumes autonomous patrol.

## CHAPTER 19 — ALGORITHMS & MATHEMATICAL FOUNDATION

### 19.1 Algorithms Actually Named or Implied in the Source Material

| Name                                       | Purpose                                                                                                    | Position in Architecture              | Status                                                             |
|--------------------------------------------|------------------------------------------------------------------------------------------------------------|---------------------------------------|--------------------------------------------------------------------|
| YOLOv8 object detection                    | Real-time visual anomaly/object detection                                                                  | Patrol layer (Chapter 10)             | Named directly in source [9]                                       |
| Weighted sensor-fusion confidence scoring  | Combine independent sensor confidences into one anomaly score                                              | Fusion layer (Chapter 14)             | Named directly in source; exact formula/weights not specified      |
| Cryptographic hashing (content addressing) | Generate a content identifier (CID) that changes if the underlying file changes, enabling tamper detection | Evidence integrity layer (Chapter 22) | Named directly in source (IPFS CIDs)                               |
| Append-only smart-contract ledger logic    | Prevent editing/deletion of stored evidence references                                                     | Evidence integrity layer (Chapter 22) | Named directly in source (no edit/delete function in the contract) |

No SLAM, path-planning, obstacle-avoidance, or ROS-based algorithm has been added here, consistent with Chapters 9 and 12.

### 19.2 Relevant Equations

Non-Maximum Suppression relies on the Intersection over Union metric:



$$IoU = \text{Area of Intersection} / \text{Area of Union}$$

This is general technical background for how YOLOv8 (Chapter 10) resolves duplicate detections, not a measured project value.

Weighted multi-sensor fusion (Chapter 14), as the Proposed Sensor Fusion Strategy:

$$C_{fused} = \sum (w_i \times C_i), \text{ subject to } \sum w_i = 1$$

Standard classification metrics (defined for future use once the team has measured data, Chapter 10.4):

$$\text{Precision} = TP / (TP + FP)$$

$$\text{Recall} = TP / (TP + FN)$$

$$F1 = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$$

Cryptographic hash one-way property (general technical background explaining the "tamper detection without tamper recovery" limitation the team states in Section 6.3 of the original report): a cryptographic hash function H is designed such that recovering the original input file F from H(F) alone is computationally infeasible; this is precisely why the source material notes that destroyed redundant copies cannot be reconstructed from the CID alone.

## CHAPTER 20 — IMPLEMENTATION STATUS

| Component                                                        | Status                                 | Evidence in Source Material                                                                                    | Remaining Work (if any)                                                                      |
|------------------------------------------------------------------|----------------------------------------|----------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| Sensor selection (camera, mmWave radar, BME688, load cell+HX711) | [PROPOSED — design finalised]          | Explicitly named and justified with rejection-of-alternatives reasoning (Chapters 4–5)                         | Physical integration and field calibration status not stated                                 |
| YOLOv8/OpenCV patrol vision pipeline                             | [PROPOSED]                             | Named as the patrol-layer detection framework [9]                                                              | Training dataset, model performance, and on-device deployment status not stated              |
| UV-fluorescence black-box confirmation module                    | [PROPOSED — chemistry-grounded design] | Detailed swab-and-chamber design described, grounded in cited literature [2,3,4]                               | Physical build/testing status not stated                                                     |
| IPFS evidence storage (Pinata + secondary pinning)               | [STATUS REQUIRES TEAM CONFIRMATION]    | Explicitly described as the prototype's storage design, but not stated whether exercised with live sensor data | Confirm whether this has been exercised end-to-end with live sensor data                     |
| Ethereum smart contract (Sepolia testnet)                        | [IMPLEMENTED — prototype, testnet]     | "deployed on the Sepolia testnet for our prototype" — explicit statement of a built artifact                   | Migration to a permissioned blockchain for production is listed as future scope (Chapter 27) |
| Offline resilience (local encrypted buffering)                   | [PROPOSED]                             | Described as part of the architecture                                                                          | Implementation/testing status not stated                                                     |
| Wheeled chassis with servo-actuated struts                       | [PROPOSED — design finalised]          | Described with clear rationale (Section 2.6)                                                                   | Physical build status not stated                                                             |
| Navigation / path planning                                       | [NOT SPECIFIED IN SOURCE MATERIAL]     | No description in source material                                                                              | Full definition required from team                                                           |
| ROS/ROS 2                                                        | [NOT APPLICABLE]                       | Not mentioned in source material                                                                               | N/A                                                                                          |
| GPS-based evidence tagging                                       | [STATUS REQUIRES TEAM CONFIRMATION]    | Implied by "GPS" in the evidence-package description (Chapter 22); module and fallback behaviour not named     | Confirm module used and indoor/tunnel fallback method (Section 12.1)                         |

| Component                               | Status     | Evidence in Source Material             | Remaining Work (if any)                          |
|-----------------------------------------|------------|-----------------------------------------|--------------------------------------------------|
| Encryption of buffered offline evidence | [PROPOSED] | Explicitly stated as a design behaviour | Confirm specific algorithm/implementation status |

Overall reading: the source material most clearly documents a built and testnet-deployed evidence-integrity (blockchain) component, alongside a thoroughly reasoned and justified sensor-selection and system design. It does not state a completed, integrated physical robot build, sensor calibration results, or a trained/evaluated vision model. This report has deliberately avoided upgrading any of these to "implemented" beyond what the source material states.

## CHAPTER 21 — TESTING METHODOLOGY (PROPOSED)

The source material does not describe a testing plan that has already been executed. The strategy below is therefore presented as a Proposed Testing Methodology for the team to execute, not as reported results.

| Test Category               | What It Would Verify                                                                                                                                                                               |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Unit testing                | Each sensor driver and the YOLOv8 inference module in isolation                                                                                                                                    |
| Sensor testing              | Response and reliability of camera, mmWave radar, BME688, and load cell+HX711 under controlled stimuli                                                                                             |
| Computer vision testing     | Detection accuracy of the YOLOv8 pipeline against a labelled validation set (once available)                                                                                                       |
| Fusion testing              | Behaviour of the weighted confidence model under sensor agreement, disagreement, missing-sensor, and noisy-sensor conditions                                                                       |
| Confirmation-module testing | Fluorescence response repeatability of the black-box module against known simulants and known negatives                                                                                            |
| Evidence-integrity testing  | Correct CID generation, correct append-only contract behaviour, correct linking of unconfirmed→confirmed records, and correct local-buffer-then-upload behaviour under simulated connectivity loss |
| Integration testing         | End-to-end flow from visual flag through fused confidence, RPF alert, confirmation, and evidence logging                                                                                           |
| Failure testing             | Behaviour under camera failure, radar failure, gas-sensor failure, load-cell failure, communication loss, and low battery                                                                          |

### 21.1 Experimental Metrics To Be Measured

No accuracy, precision, recall, mAP, FPS, detection range, battery life, navigation success rate, sensor sensitivity, response time, latency, or power-consumption figure is stated in the source material, and none has been invented for this report. The table below lists the metrics a jury or faculty evaluator would expect, together with the methodology the team should use to measure each one before citing a number.

| Metric                           | Definition/Reference | Suggested Measurement Methodology                                                                                                                            |
|----------------------------------|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Precision / Recall / F1 (vision) | Chapter 10.4         | Run YOLOv8 inference on a held-out, labelled validation set of candidate-object images; count true/false positives and negatives against ground-truth labels |
| mAP@0.5 / mAP@0.5:0.95           | Chapter 10.4         | Standard COCO-style evaluation protocol against the labelled validation set, once one exists                                                                 |

| Metric                                               | Definition/Reference                      | Suggested Measurement Methodology                                                                                                                                                                                                                                         |
|------------------------------------------------------|-------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Detection range (camera / radar)                     | N/A — no figure stated in source material | Place calibrated test objects at measured distances under controlled lighting/RF conditions; record the maximum distance at which each sensor reliably registers the object; report as an experimentally validated range, distinct from any manufacturer datasheet figure |
| Fusion-layer false-alarm rate                        | Chapter 14                                | Run controlled trials with known-negative objects varying in shape, weight, and VOC background; record how often the fused score crosses the escalation threshold                                                                                                         |
| Confirmation-module sensitivity/specificity          | Section 5.3                               | Test the black-box module against a panel of legal simulants and known-negative swabs; compute true/false positive and negative rates                                                                                                                                     |
| End-to-end latency (flag to fused confidence)        | Chapter 18                                | Timestamp each pipeline stage during controlled runs; report mean/percentile latency from camera flag to fused confidence output                                                                                                                                          |
| Patrol endurance / battery life                      | N/A — battery not specified (Chapter 7)   | Run a continuous patrol-mode discharge test from full charge to cutoff under representative duty cycle                                                                                                                                                                    |
| Evidence-logging reliability under connectivity loss | Section 22.1                              | Simulate network outages of varying duration; verify zero evidence loss and correct automatic upload on reconnection                                                                                                                                                      |

Until these are measured, any performance claim about RAKSHAK K9 should be stated as a design target or expected range, not as a validated result.

## CHAPTER 22 — SAFETY, FAILURE HANDLING & RELIABILITY

### 22.1 Evidence Integrity: IPFS and Blockchain

#### Why Not a Conventional Database

A conventional database is controlled by whoever administers it — records can be silently altered or deleted with no way to prove tampering occurred. For evidence intended to support a legal or disciplinary process, this is identified in the source material as a critical weakness.

#### Architecture

- IPFS (via Pinata, with a secondary independent pinning provider for redundancy) stores the actual evidence files (images, sensor readings, GPS, timestamps), and returns a content hash (CID) that changes completely if the file is altered in any way.
- An Ethereum smart contract (deployed on the Sepolia testnet for the prototype) stores only the CID and metadata, in an append-only structure — there is no edit or delete function in the contract, making tampering with a stored record structurally impossible rather than merely policy-restricted.
- Evidence is logged in two linked stages: an "unconfirmed" package generated the moment sensors flag an anomaly, and a "confirmed" package generated after RPF's black-box verification, referencing the original record's CID/transaction ID. This preserves the full investigation timeline immutably, rather than only the final outcome.

- If network connectivity is unavailable (e.g., remote yards, tunnels), evidence is encrypted and buffered locally, then automatically uploaded once connectivity resumes.

### Honest Limitation (as stated in the source material)

This design guarantees tamper detection: any alteration to stored evidence is immediately provable via hash mismatch. It does not guarantee tamper recovery: if all redundant copies of a file were somehow destroyed simultaneously, the original content cannot be reconstructed from the hash alone, since cryptographic hashes are one-way functions by design.

## 22.1a Cybersecurity Mechanism Status

Security-relevant mechanisms must be reported individually rather than as a single "the system is secure" claim. The table below classifies each mechanism named or implied in the source material.

| Mechanism                                                           | Category                       | Status                                     | Basis                                                                                    |
|---------------------------------------------------------------------|--------------------------------|--------------------------------------------|------------------------------------------------------------------------------------------|
| Local evidence encryption during offline buffering                  | Encryption                     | [PROPOSED]                                 | Explicitly stated: "evidence is encrypted and buffered locally" during connectivity loss |
| Content-addressed hashing (IPFS CID)                                | Integrity                      | [IMPLEMENTED — prototype]                  | CID generation is inherent to the stated IPFS storage design                             |
| Append-only smart-contract record (no edit/delete function)         | Integrity / audit logging      | [IMPLEMENTED — prototype, Sepolia testnet] | Explicitly stated as deployed for the prototype                                          |
| Redundant IPFS pinning (Pinata + secondary provider)                | Integrity (availability)       | [PROPOSED]                                 | Described as part of the architecture; operational status not confirmed                  |
| Authentication of RPF personnel / access control to confirm records | Authentication / Authorization | [STATUS REQUIRES TEAM CONFIRMATION]        | Not described in source material                                                         |
| Transport-layer security (e.g., TLS) for data in transit            | Encryption (in transit)        | [NOT SPECIFIED IN SOURCE MATERIAL]         | No transport security mechanism named                                                    |
| Secure/OTA firmware or software update mechanism                    | Secure update                  | [NOT SPECIFIED IN SOURCE MATERIAL]         | No update mechanism named                                                                |
| Command validation / unauthorized command prevention                | Authorization                  | [NOT SPECIFIED IN SOURCE MATERIAL]         | No remote command interface is described in enough detail to assess this                 |

Encryption, authentication, authorization, integrity, secure update, and audit logging are distinct properties; none should be assumed present merely because the architecture is described as using "blockchain" or "encryption" in general terms. Only the two mechanisms above marked [IMPLEMENTED — prototype] have an explicit statement in the source material of having been built and deployed (to a testnet, in the blockchain case).

## 22.2 Failure Modes Not Otherwise Specified

[NOT SPECIFIED IN SOURCE MATERIAL: emergency-stop mechanism, collision-avoidance behaviour, safe-velocity limiting, battery-protection behaviour, and sensor-failure fallback behaviour beyond the offline evidence-buffering case above.] These are flagged as gaps the team should address in a future revision of the project documentation, given their importance to a physical robot operating in a public railway environment.

## 22.3 Human-in-the-Loop as a Reliability Mechanism

Because every consequential action (RPF alert, confirmation) requires human involvement (Chapter 15.3), the overall system's reliability with respect to false action is bounded by human review rather than resting entirely on sensor/software reliability — a design property the source material presents as intentional.

## CHAPTER 23 — RISK ANALYSIS

The matrix below is the report's engineering construction, built from limitations and design statements the source material makes explicitly (Chapters 4, 22, 25); it is labelled conceptual where the source does not quantify probability or severity.

| Risk                                              | Cause                                                                                           | Effect                                                                                                | Probability (qualitative)             | Severity (qualitative) | Mitigation (as documented or proposed)                                                                                       |
|---------------------------------------------------|-------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|---------------------------------------|------------------------|------------------------------------------------------------------------------------------------------------------------------|
| False negative (missed threat)                    | Non-volatile, non-specific, or novel substances evade all patrol-layer sensors (Chapter 4)      | Threat passes undetected                                                                              | Moderate (inherent to sensor physics) | High                   | Multi-sensor fusion raises detection probability versus any single sensor; human RPF oversight remains the ultimate backstop |
| False positive (false alarm)                      | Benign object matches visual/weight/gas anomaly pattern                                         | Unnecessary RPF response, potential public disruption                                                 | Moderate                              | Low-Moderate           | Confirmation layer requires human-verified fluorescence test before "confirmed" status is logged                             |
| Evidence tampering attempt                        | Malicious alteration of stored evidence                                                         | Compromised legal chain of custody                                                                    | Low (given cryptographic design)      | High                   | IPFS content-addressing + append-only smart contract make tampering detectable via hash mismatch                             |
| Catastrophic loss of all evidence copies          | Simultaneous failure/deletion of all IPFS pinning copies                                        | Original evidence content unrecoverable (tamper detection ≠ tamper recovery, explicitly acknowledged) | Low                                   | High                   | Secondary independent pinning provider for redundancy; full recovery guarantee not claimed                                   |
| Network connectivity loss in remote yards/tunnels | Physical/RF coverage gaps                                                                       | Delayed evidence logging                                                                              | Moderate (railway yards/tunnels)      | Low-Moderate           | Local encrypted buffering with automatic upload on reconnection                                                              |
| Public/production use of a public blockchain      | Ethereum mainnet-style public visibility and transaction cost do not suit production deployment | Evidence records could be publicly visible; costs would not scale across a fleet                      | Certain if unaddressed                | Moderate               | Migration to a permissioned blockchain (e.g., Hyperledger Fabric) identified as required future work                         |
| Sensor/hardware failure in field                  | Environmental exposure, wear, power issues                                                      | Reduced screening coverage or missed detections                                                       | [Not quantified in source material]   | Moderate-High          | [Fallback behaviour not specified in source material — identified gap]                                                       |

## CHAPTER 24 — NOVELTY / INNOVATION AND SIH RELEVANCE

Based on the team's market survey (Chapter 3), the source material states that no existing commercial product or publicly documented research prototype was found that combines all of the following in a single mobile system:

1. Autonomous multi-sensor patrol robot (not fixed, not purely handheld);
2. Weighted sensor-fusion confidence scoring across independent physical detection principles;
3. An on-board, low-cost, human-verified fluorescence confirmation stage for suspected explosive residue;
4. A two-stage (unconfirmed → confirmed), cryptographically linked, IPFS + blockchain evidence trail with offline resilience.

The team frames this explicitly as its novelty claim based on the literature and commercial products it was able to find during research, while noting candidly that it has not been able to review every unpublished or proprietary system that may exist, and recommends the claim be independently verified against the SIH evaluation panel's own domain knowledge before being presented as an absolute statement. This report preserves that qualification rather than presenting the novelty claim as an unconditional "first ever" statement, per the source material's own stated caution.

Relevance to SIH26026: the project directly addresses the sponsoring ministry's stated need for a mobile detection system, offers a wheeled alternative to the example quadruped/handheld form factors with an explicit engineering rationale (Section 2.6), and adds an evidence-integrity dimension (legally defensible chain of custody) that is not part of the literal problem-statement text but that the team identifies as a differentiator relative to existing mobile security-robot products (Chapter 3).

## CHAPTER 25 — COMPARISON WITH EXISTING APPROACHES

| Parameter                                    | Manual/Canine Inspection                         | Fixed Installation (X-ray/IMS/Raman/NQR)                                                     | Generic Patrol/Security Robot                                                     | RAKSHAK K9 (Proposed)                                                                                                 |
|----------------------------------------------|--------------------------------------------------|----------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| Human exposure to unscreened areas           | High (limited hours/coverage)                    | Low at checkpoint, but areas outside checkpoint remain unscreened                            | Low (visual surveillance only)                                                    | Reduced via continuous autonomous first-pass screening                                                                |
| Mobility / coverage                          | Limited by personnel/dog working hours           | None — fixed location                                                                        | Mobile, but visual-surveillance only                                              | Mobile, multi-sensor                                                                                                  |
| Detection modality                           | Trained olfactory (canine) or manual inspection  | Single lab-grade modality (X-ray, IMS, Raman, or NQR)                                        | Typically visual/thermal only                                                     | Fused camera + radar + gas + weight, plus chemistry-grounded confirmation                                             |
| Legally defensible evidence chain            | Dependent on manual documentation                | Dependent on facility's own record-keeping (source material does not compare this in detail) | Not typically integrated with a tamper-proof ledger, per the team's market survey | IPFS + blockchain two-stage evidence trail (Chapter 22)                                                               |
| Approximate cost per unit                    | High recurring cost (training, handler, welfare) | \$15,000–\$100,000+ (Chapter 3)                                                              | \$5,000–\$30,000+ (Chapter 3)                                                     | Designed around low-cost sensors, though the team's own total unit cost is not stated in the source material          |
| Scalability to continuous, unmanned coverage | Low                                              | Low (fixed locations only)                                                                   | Moderate (mobile, but detection capability limited)                               | Positioned by the team as its target advantage, though not independently verified against production-scale deployment |

## CHAPTER 26 — LIMITATIONS

The following limitations are drawn directly from statements and implications in the source material, not invented for this report:

- Physical detection constraint: most narcotics and several explosives (RDX, PETN) are non-volatile, limiting the effectiveness of any vapour/VOC-based sensing as a standalone method (Chapter 4).
- Non-specificity of low-cost sensors: the gas/VOC sensor responds to VOCs broadly, not to a specific compound; the load-cell heuristic and visual detection are proxies, not chemical confirmations (Chapter 14.5).
- Human-dependency of the confirmation stage: the black-box module requires RPF personnel to physically collect a swab and apply reagent; it is not a fully autonomous confirmation step, by design (Section 5.4).
- Tamper detection without tamper recovery: the evidence-integrity layer can prove tampering occurred but cannot reconstruct destroyed content from a hash alone if all redundant copies are lost (Section 22.1).
- Public-blockchain suitability: the current Ethereum Sepolia testnet approach is explicitly a prototype choice; the source material states that a public blockchain is not suited to production due to evidence visibility and transaction-cost scaling concerns (Chapter 26 in the original report / Chapter 27 of this report).
- Terrain scope: the wheeled, servo-actuated design targets predominantly flat, paved railway surfaces; the source material's own future-scope section acknowledges a gap toward more extreme terrain (ballast sections, stairs) relative to the quadruped example in the problem statement (Section 2.6, Chapter 27).
- Training-data scope: any vision-model training to date has reportedly used legal simulants rather than authorised real narcotics/explosives samples, per the future-scope statement about expanding data via authorised agencies (Chapter 27).
- Unquantified performance: the source material provides no measured detection accuracy, false-alarm rate, patrol endurance, or end-to-end latency figures; this report has not fabricated any such figures.
- Undocumented navigation and safety subsystems: as detailed in Chapters 12, 16, and 22, several subsystems typically expected in an autonomous mobile robot (navigation/path-planning method, emergency stop, collision avoidance) are not described in the source material and require clarification from the team.

## CHAPTER 27 — FUTURE SCOPE

As documented in the source material:

- Integration of lab-grade confirmation hardware (portable IMS or NQR modules, as they become more affordable, e.g., following the cost-reduction trajectory demonstrated by low-cost portable NQR research [7]) as a drop-in replacement or supplement to the black-box fluorescence stage.
- Migration to a permissioned blockchain (e.g., Hyperledger Fabric) for production deployment, since evidence should not be publicly visible the way a public Ethereum record is, and transaction costs would not scale economically across a large fleet.
- Fleet coordination: multiple RAKSHAK K9 units patrolling different zones of a station or yard, coordinated through a shared control dashboard.
- Mobility upgrade path: the servo-actuated ground-clearance mechanism could be extended toward a hybrid wheel-leg or tracked design for more extreme terrain (ballast sections, stairs), narrowing the gap toward the quadruped form factor referenced in the original problem statement.
- Expanded AI training data: partnering with authorised agencies (RPF, CISF, forensic labs) to legally and safely expand the training dataset for both the vision model and the fluorescence-confirmation classifier, beyond the legal simulants used during prototype development.
- Solar-assisted power system to extend patrol duration for remote or off-grid railway sections.

This report additionally flags, as candidate future work arising from gaps identified during this documentation exercise (not as claims already made by the team): formal specification of a navigation/path-planning approach, a documented emergency-stop and collision-avoidance scheme, a defined and calibrated fusion-weight methodology, and a structured field-testing and performance-measurement plan (Chapter 21).

## CHAPTER 28 — IMPLEMENTATION ROADMAP (PROPOSED)

The following phased roadmap is the report's own construction, sequencing the components the source material describes into a realistic build-out order for a student SIH team. It is explicitly proposed/illustrative, since the source material does not itself present a phased roadmap.

| Phase                                               | Objective                                                              | Key Tasks                                                                                                  | Validation Criteria                                                                           |
|-----------------------------------------------------|------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| Phase 1 — Chassis & mobility                        | Build the wheeled platform with servo-actuated struts                  | Assemble chassis, wheels, struts, servo control; verify ground-clearance adjustment                        | Robot can move and adjust ground clearance on a flat test surface                             |
| Phase 2 — Patrol-layer sensing                      | Integrate the always-on camera and vision pipeline                     | Mount camera; deploy YOLOv8/OpenCV; verify continuous low-power operation                                  | Camera streams continuously; YOLOv8 produces detections on test objects                       |
| Phase 3 — Inspection-layer sensing                  | Integrate mmWave radar, BME688, load cell + HX711                      | Wire and power sensors; verify activation-on-flag behaviour                                                | Inspection sensors correctly power on only after a visual flag and return readings            |
| Phase 4 — Fusion logic                              | Implement weighted confidence scoring                                  | Define initial weights; implement Section 14.3 model; test against known scenarios                         | Fusion output responds sensibly to simulated sensor agreement/disagreement                    |
| Phase 5 — Confirmation module                       | Build and validate the UV-fluorescence black-box                       | Assemble sealed chamber, UV emitter, photosensor; test with legal simulants and known negatives            | Module reliably distinguishes treated-positive vs. negative swabs in controlled tests         |
| Phase 6 — Evidence integrity                        | Deploy IPFS + smart-contract pipeline end-to-end with real sensor data | Connect Pinata + secondary pinning; verify Sepolia contract read/write; test unconfirmed→confirmed linking | A full detection event produces a correctly linked, retrievable evidence pair on IPFS/testnet |
| Phase 7 — Offline resilience                        | Implement and test local buffering                                     | Simulate connectivity loss; verify encrypted local storage and auto-upload on reconnection                 | No evidence loss across a simulated connectivity outage                                       |
| Phase 8 — Integration & field testing               | Run the full pipeline together on the physical robot                   | End-to-end test per Chapter 21                                                                             | Successful end-to-end run from patrol through confirmed evidence logging                      |
| Phase 9 — Documentation of navigation/safety gaps   | Close the gaps identified in Chapters 12, 16, 22                       | Specify and implement navigation method, emergency-stop, and failure fallback behaviour                    | Documented and demonstrable safety behaviour under at least one failure scenario              |
| Phase 10 — Roadmap toward production (future scope) | Address Chapter 27 future-scope items                                  | Permissioned blockchain migration, fleet coordination, expanded training data, solar power                 | Per the team's own future-scope criteria; not required for SIH prototype demonstration        |

## CHAPTER 29 — CONCLUSION



RAKSHAK K9 is designed around a principle the team states it arrived at only after evaluating and rejecting several single-sensor approaches: no affordable sensor reliably identifies explosives or narcotics on its own, and no currently available product combines autonomous mobile patrol with a legally defensible, tamper-proof evidence trail. Rather than overclaiming detection certainty, the system is positioned honestly as a continuous, low-cost, multi-sensor screening and evidence-logging layer that raises the probability of catching genuine threats through independent, fused signals, escalates credible flags to RPF for human-verified confirmation, and preserves an immutable record of the entire process from first detection through final confirmation.

This report has expanded the team's original Detailed Project Report into chapter form for technical evaluation while deliberately declining to add unsupported hardware specifications, ROS architecture, navigation algorithms, dataset statistics, or performance figures. The clearest next step for strengthening the project ahead of jury evaluation is closing the gaps flagged throughout this report — particularly navigation/path-planning method, safety/fail-safe behaviour, fusion-weight calibration, and a first round of measured testing results (Chapters 12, 16, 21, 22, 25) — since these are the areas most likely to be probed in technical viva.

## CHAPTER 30 — TECHNICAL VALIDATION & CLAIM AUDIT

This chapter documents the validation pass applied to the source material before and during the writing of this report. It exists so the jury and the team can see explicitly which claims were tightened, which were already technically defensible, and which remain open.

### 30.1 Validation Checklist Applied

| # | Validation Area                                                   | Outcome                                                                                                                                                                                                                                                                                                                                                                                              |
|---|-------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Sensor capability (BME688 not presented as a definitive detector) | Applied — reworded to "VOC-anomaly screening"; confirmatory mechanism identified explicitly (Section 5.2, 14.1a)                                                                                                                                                                                                                                                                                     |
| 2 | Detection-range claims                                            | No numeric detection range (e.g., "30–60 cm") is stated anywhere in the source material, so none appears in this report; if the team later supplies one, it must be labelled manufacturer specification / design target / expected range / experimentally validated range as appropriate                                                                                                             |
| 3 | AI vs chemical detection separation                               | Applied — Section 14.1a taxonomy (A–F) keeps visual, spatial, VOC, confirmatory-chemical, and final-classification evidence separate                                                                                                                                                                                                                                                                 |
| 4 | Terminology                                                       | Reviewed for absolute claims ("100% detection," "guaranteed," "definitive identification"); none were found in the drafted text beyond the verbatim official problem-statement title and correctly-scoped statements about the append-only ledger's tamper-detection guarantee, which the source material itself states and this report qualifies with its tamper-recovery limitation (Section 22.1) |
| 5 | Hardware naming ambiguity                                         | No informal/ambiguous component names (e.g., "Pi," "Nano") appear in the source material; all unnamed components are marked "Specification not provided in source material" (Chapter 7.1)                                                                                                                                                                                                            |
| 6 | Quadruped/autonomy vs operator control                            | Applied — Section 15.4 separates autonomous functions from operator-assisted (RPF) functions explicitly                                                                                                                                                                                                                                                                                              |
| 7 | ROS/ROS 2                                                         | Applied — Chapter 9 states no ROS/ROS2 element is evidenced; nothing invented                                                                                                                                                                                                                                                                                                                        |
| 8 | SLAM                                                              | Applied — Chapter 12 states no SLAM implementation is named; no toolbox (SLAM Toolbox, Cartographer, RTAB-Map, GMapping, Nav2 SLAM) is claimed                                                                                                                                                                                                                                                       |
| 9 | Navigation subsystem separation                                   | Applied — Chapter 12 separates localisation, mapping, planning, obstacle avoidance, and control, marking each not specified rather than assuming one sensor performs all functions                                                                                                                                                                                                                   |

| #  | Validation Area                       | Outcome                                                                                                                                                                                  |
|----|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 10 | Camera-gated sensor activation        | Applied — Section 5.2 describes it as a gating mechanism with intended (not measured) benefits                                                                                           |
| 11 | Blockchain claims                     | Applied — "tamper-evident"/"integrity-verifiable" used; blockchain explicitly stated to store only CIDs/metadata, not raw evidence files (Section 22.1, 22.1a)                           |
| 12 | IPFS claims                           | Applied — IPFS's role limited to content-addressed storage/reference; confidentiality/authentication not attributed to IPFS itself (Section 22.1a)                                       |
| 13 | Cybersecurity mechanism separation    | Applied — new Section 22.1a separates encryption, authentication, authorization, integrity, secure update, and audit logging with individual status labels                               |
| 14 | GPS availability                      | Applied — new Section 12.1 states indoor/tunnel/rail-yard GPS limitations and flags the missing fallback method                                                                          |
| 15 | Dataset validation                    | Applied — new Section 10.3a separates dataset categories A–E and states none is confirmed as a validated narcotics/explosives training set                                               |
| 16 | Experimental result fabrication       | Applied — no accuracy/precision/recall/mAP/FPS/range/battery/latency/power figures invented anywhere in this report; Section 21.1 "Experimental Metrics To Be Measured" provided instead |
| 17 | Implemented/Proposed/Future labelling | Applied — Chapter 20 and Section 22.1a use bracketed [IMPLEMENTED]/[PROPOSED]/[STATUS REQUIRES TEAM CONFIRMATION]/[NOT SPECIFIED IN SOURCE MATERIAL] labels consistently                 |
| 18 | Claim-strength review                 | Applied — see audit table below (Section 30.2) for the specific claims reviewed and, where needed, restated                                                                              |

## 30.2 Final Claim Audit Table

| Claim (as it might appear in a pitch/PPT) | Source Evidence                                                                      | Technical Risk                                                                                                                | Correction Applied in This Report                                                                                                        | Status    |
|-------------------------------------------|--------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| "BME688 detects narcotics/explosives"     | BME688 named as a gas/VOC sensor in the sensor-selection section                     | High — implies compound-specific chemical identification the sensor cannot provide (Chapter 4.2)                              | Restated as "VOC-anomaly screening input"; confirmatory mechanism identified separately (Section 5.2, 14.1a)                             | Corrected |
| "AI detects explosives"                   | YOLOv8 named as the patrol-layer vision framework                                    | High — merges visual candidate-detection with chemical threat identification                                                  | Restated across Chapter 10/14.1a: vision provides candidate/visual evidence only, not chemical identification                            | Corrected |
| "Blockchain makes evidence tamper-proof"  | Source states append-only contract with no edit/delete function                      | Moderate — "tamper-proof" overstates the guarantee (tamper detection ≠ tamper recovery, which the source itself acknowledges) | Restated as "tamper-evident"/"integrity-verifiable," with the explicit tamper-detection-vs-recovery distinction preserved (Section 22.1) | Corrected |
| "The robot provides real-time detection"  | Source describes continuous camera monitoring and inspection-stage sensor activation | Moderate — "real-time" implies a measured latency bound that is not provided                                                  | Restated as near-real-time/continuous monitoring by design; latency identified as                                                        | Corrected |

| Claim (as it might appear in a pitch/PPT)                                                      | Source Evidence                                                                                                                          | Technical Risk                                                            | Correction Applied in This Report                                                                                                       | Status                     |
|------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|----------------------------|
|                                                                                                |                                                                                                                                          |                                                                           | unmeasured (Section 21.1)                                                                                                               |                            |
| "Novelty: first system of its kind"                                                            | Source's own novelty section already self-qualifies this claim                                                                           | Low — source material itself already hedges appropriately                 | Preserved the source's own qualification rather than strengthening it (Chapter 24)                                                      | Already defensible         |
| "Cryptographic evidence chain is unbreakable"                                                  | Source explicitly states tamper detection without tamper recovery, and one-way hash property                                             | Low — source already states the honest limitation                         | Preserved verbatim limitation; added general hash one-way-function explanation as background (Chapter 19.2)                             | Already defensible         |
| "Wheeled robot is a full substitute for the quadruped form factor in the problem statement"    | Source gives an engineering rationale for wheels on flat, paved terrain, and separately acknowledges a terrain gap toward ballast/stairs | Moderate — could be read as claiming full terrain parity with a quadruped | Both the rationale and the acknowledged terrain gap are presented together (Section 2.6, Chapter 26, Chapter 27)                        | Corrected/balanced         |
| "Continuous GPS-tagged evidence"                                                               | GPS mentioned only as part of the evidence package                                                                                       | Moderate — implies GPS is always available in the target environment      | Added explicit indoor/tunnel/rail-yard GPS limitation and flagged missing fallback method (Section 12.1)                                | Corrected                  |
| "System uses AES-256/TLS/PKI/RBAC" (not stated in source, hypothetical risk for future drafts) | Not present in source material                                                                                                           | High if added without evidence                                            | No such claim has been added; Section 22.1a instead lists only the two mechanisms the source material actually evidences as implemented | Avoided — flagged for team |

### 30.3 Items Still Requiring Team Confirmation

[DETAIL REQUIRED FROM TEAM]: exact compute unit/SBC or microcontroller; motor driver and battery specification; GPS module and its offline fallback method; patrol route/navigation method (pre-programmed, teleoperated, or autonomous); emergency-stop and collision-avoidance mechanism; exact fusion weights and escalation threshold; whether the IPFS/blockchain pipeline has been tested end-to-end with live sensor data; authentication/access-control method for confirming evidence records; and the composition of any dataset used to train the vision model to date.

## REFERENCES

### A. External Technical References (from the source material's bibliography)

- [1] Cooper, J. K.; Grant, C. D.; Zhang, J. Z. Experimental and TD-DFT Study of Optical Absorption of Six Explosive Molecules: RDX, HMX, PETN, TNT, TATP, and HMTD. *J. Phys. Chem. A* 2013, 117, 6043–6051. DOI: 10.1021/jp312492v.
- [2] Yang, J.-S.; Swager, T. M. Fluorescent Porous Polymer Films as TNT Chemosensors: Electronic and Structural Effects. *J. Am. Chem. Soc.* 1998, 120(46), 11864–11873. DOI: 10.1021/ja982293q.
- [3] Narayanan, A.; Varnavski, O. P.; Swager, T. M.; Goodson, T. Multiphoton Fluorescence Quenching of Conjugated Polymers for TNT Detection. *J. Phys. Chem. C* 2008, 112(4), 881–884. DOI: 10.1021/jp709662w.

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- [5] Smith, J. A. S. Nitrogen-14 Quadrupole Resonance Detection of RDX and HMX Based Explosives. *European Convention on Security and Detection* 1995, 408, 288–292.
- [6] Somasundaram, S. D.; Althoefer, K.; Smith, J. A. S.; Seneviratne, L. D. Detection of Landmines Using Nuclear Quadrupole Resonance (NQR): Signal Processing to Aid Classification. In *Climbing and Walking Robots*; Springer: Berlin, Heidelberg, 2006. DOI: 10.1007/3-540-26415-9\_100.
- [7] Otagaki, Y.; Miyato, Y.; Barras, J.; et al. Development of a Low-Cost, Portable NQR Spectrometer for RDX Explosives Detection. *IEEE Sensors Journal* 2020.
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- [9] Jocher, G.; Chaurasia, A.; Qiu, J. Ultralytics YOLOv8. Software, Ultralytics, 2023. <https://github.com/ultralytics/ultralytics>.

## B. Information Obtained From Uploaded Project Material

All sensor selections, market-survey figures, the two-layer architecture, the black-box fluorescence module design, the IPFS/blockchain evidence-integrity design, the novelty statement, and the future-scope list in this report are drawn directly from the team's original Detailed Project Report (RAKSHAK K9, SIH26026).

## C. Engineering Interpretation / Inference Added in This Report

The layered system-architecture framing (Chapter 6), the conceptual threat-decision matrix (Chapter 15.2), the risk-analysis matrix (Chapter 23), the proposed testing methodology (Chapter 21), and the proposed implementation roadmap (Chapter 28) are this report's own structuring of the source material for jury-readability, clearly labelled as conceptual/proposed rather than as verbatim project claims.

Note on cost figures (restated from the source material): cost figures for commercial detection systems in Chapter 3 are approximate ranges compiled from publicly available product literature and industry reporting at the time of the original report's writing; exact pricing is typically available only on direct enquiry to manufacturers and was not independently verified through a formal quote process.

# APPENDIX A — VERBATIM PROBLEM STATEMENT

Problem Statement ID: SIH26026. Theme: Robotics and Drones. Category: Hardware. Sponsoring Ministry: Ministry of Railways. Statement: "Development of Mobile (Quadruped)/Handheld Device/System for Real-Time Detection of Narcotics and Explosives across Indian Railways."

# APPENDIX B — CONSISTENCY / QUALITY-CONTROL CHECKLIST

The following checklist was applied before finalising this report: project name consistent (RAKSHAK K9) — yes; problem statement unaltered — yes; hardware list consistent with source (camera, mmWave radar, BME688, load cell+HX711, UV emitter+photosensor, servo-actuated struts, 4-wheel chassis) — yes; software list consistent (YOLOv8, OpenCV, IPFS/Pinata, Ethereum/Sepolia) — yes; no ROS terminology or nodes fabricated — confirmed; no navigation/SLAM architecture fabricated — confirmed; implemented/proposed/future components separated (Chapter 20) — yes; no unsupported performance numbers introduced — confirmed; no fake specifications introduced (gaps marked "not specified in source material") — confirmed; no fake datasets or experimental results introduced — confirmed; equations limited to those relevant to named technologies (Chapter 19) — confirmed; diagrams match the written architecture — confirmed; references distinguish external literature, source-material content, and this report's own interpretation (References, Sections A–C) — confirmed.